

EEE 318 (July 2024)
EEE 318: Control System I Laboratory

Final Project Report

Section: B2 Group: 03

Librarian Arm

Course Instructors:

Mrinmoy Kundu, Lecturer
Md. Meherab Hossain, Part-Time Lecturer

Signature of Instructor: _____

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Signature: _____ Full Name: Arpon Kumar Chanda Student ID: 2006110	Signature: _____ Full Name: Md. Kaidul Islam Student ID: 2006111	
Signature: _____ Full Name: Sadaf Shafin Chowdhury Student ID: 2006113	Signature: _____ Full Name: Shohan Ahmed Rifat Student ID: 2006116	
Signature: _____ Full Name: Fabliha Labiba Student ID: 2006119		

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1 Abstract

The Librarian Arm is a semi-autonomous robotic system designed to locate and retrieve a specific book from a shelf. This system leverages an ESP32 CAM module running computer vision algorithms and ARUCO marker detection, enabling the robot to identify a target book visually. Ultrasonic sensors provide distance feedback to guide a DC gear motor and a linear actuator in precise positioning. A force-sensitive resistor (FSR) ensures that the clamp securely holds the book without causing damage. Servo motors allow for the adjustment of tilt and gripper angles. The entire sequence is controlled by an Arduino MEGA 2560. Testing has demonstrated the system's ability to approach the target, adjust its position, clamp the book, and retrieve it safely and autonomously. This project lays the groundwork for more sophisticated autonomous retrieval solutions and highlights the potential of low-cost mechatronic integration in library settings.

2 Introduction

Libraries contain vast collections of books that often need to be fetched and returned frequently. Repetitive manual retrieval can be time-consuming, labor-intensive, and may pose challenges for individuals with limited mobility. The Librarian Arm project addresses these issues by integrating mechanical actuators, sensors, and computer vision into a single, coherent robotic solution.

Context and Motivation:

- Manual retrieval requires human effort and time, particularly in large libraries or archives.
- Automation can assist librarians by performing routine retrieval tasks, allowing staff to focus on assisting patrons and curating collections.

Why This Is a Complex Engineering Problem:

- The system integrates multiple engineering domains: mechanical design (actuators, clamps), electronics (drivers, converters), control systems (sensor feedback loops) and computer vision (ARUCO markers).
- Balancing cost, accuracy, durability, and safety introduces complexity.

Alternative Solutions:

- Existing library retrieval systems may rely on RFID tags or barcodes, requiring specialized scanning equipment.
- Fully robotic industrial arms are costlier and complex, and may be overkill for simple retrieval tasks.
- Our approach uses cost-effective components and open-source vision tools, striking a balance between complexity and accessibility.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The objective is to design a robotic platform capable of:

- Identifying a target book's location via ARUCO marker detection from a mounted ESP32 CAM.
- Moving a DC-driven platform to an optimal distance in front of the shelf.
- Using a linear actuator and servos to adjust vertical and angular positioning for book extraction.
- Employing a force-sensitive resistor to ensure gentle gripping and safe book handling.

3.1.2 Literature Review

Research in automated retrieval often spans from robotic arms in warehouses to automated guided vehicles and specialized library robots. ARUCO markers are commonly used in robotics for localization due to their robust, low-complexity detection. While RFID-based systems exist, they lack the direct visual feedback of camera-based methods. Our approach leverages accessible open-source technologies and standard hardware components.

3.1.3 Formulation of Problem

Key engineering considerations include:

- Determining the correct distances and angles at which to stop the motor and extend the actuator.
- Ensuring that the clamp and tilt mechanism handle the book securely without overexerting force.
- Providing robust visual detection even under varying lighting conditions.

3.1.4 Analysis

Key parameters include:

- Ultrasonic Sensor Thresholds: Determining stop and approach distances.
- FSR Threshold: Identifying a safe gripping force.
- Actuator Speed and Stroke: Ensuring sufficient travel to reach the shelf and adjust for book height.
- Timing and Sequencing: Coordinating movements for smooth operation.

3.2 Design Method (PO(a))

Mathematical and Engineering Knowledge Application:

- Distance measurements from ultrasonic sensors (HC-SR04) are used to trigger motor stops and actuator movements based on pre-defined thresholds (e.g., 13 cm, 22 cm from the shelf surface).
- The FSR provides analog feedback on applied pressure, allowing the controller to stop retracting once a safe threshold is reached.
- Basic kinematic calculations ensure that the 150 mm stroke actuator can accommodate the required vertical travel.

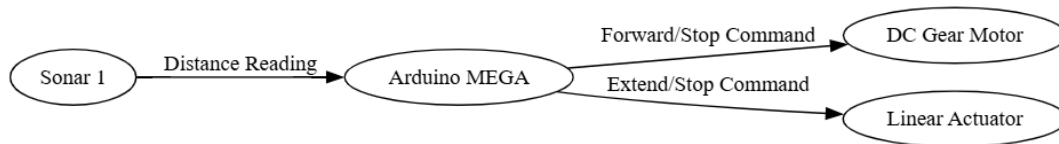
Control Logic and Feedback Loops:

- A finite state machine (FSM) approach sequences actions: open clamp, move forward, stop at distance, extend actuator, tilt, sense pressure, and so forth.
- Visual feedback from the ESP32 CAM confirms the presence of the ARUCO marker before initiating the sequence.

3.3 Circuit Diagram

- System Components:
 - **Arduino MEGA:** Central controller.
 - **ESP32 CAM:** Vision module for ARUCO marker detection.
 - **Ultrasonic Sensors (Sonar 1 & Sonar 2):** Measure distances to trigger actuator steps.

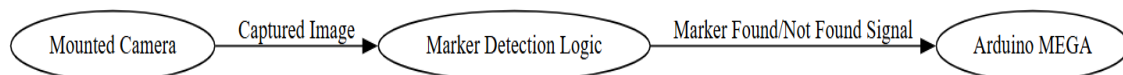
- **Pressure Sensor (FSR):** Ensures safe clamping.
- **DC Gear Motor & Linear Actuator:** Move arm in horizontal and vertical directions.
- **Servos:** Control clamp and tilt.
- **Control Logic:**
 - State machine approach: Detect ARUCO → Move Forward → Extend Actuator → Tilt → Clamp → Retract → Return.
- **Block Diagram**
 - Sonar 1 Feedback Block Diagram



- Sonar 2 Feedback Block Diagram



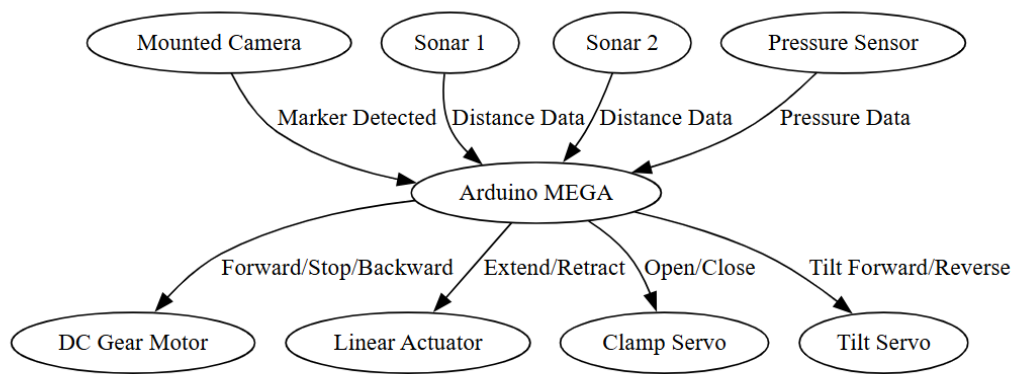
- Mounted Camera Feedback Block Diagram



- Pressure Sensor Feedback Block Diagram



- Complete Workflow Block Diagram



3.4 Simulation and Testing

- Early simulations validated:
 - Sensor response timing.
 - Distance thresholds for stopping conditions.
 - Feasibility of actuator stroke length for required vertical displacement.
- MATLAB/Simulink or similar tools could be used to model the sensor-actuator loops and verify timing before hardware tests.

(No PCB was designed or manufactured; all sensor integrations and component connections were made on a breadboard for prototyping ease.)

3.5 Full Source Code of Firmware

The Arduino firmware code is provided in the Appendix. It includes setup routines, sensor reading functions, actuator control functions, and a main loop implementing the state machine for the retrieval sequence.

4 Implementation

4.1 Workflow

4.1.1 Mechanical Subsystems:

- **DC Gear Motor (12V, 500 RPM):** Moves the entire system platform forward and backward along a linear track.
- **Linear Actuator (150 mm Stroke):** Adjusts vertical position to align with the target book's height.

- **Servos (MG996R):** One servo controls the clamp for gripping the book; another servo-based mechanism tilts the book for easier extraction.

4.1.2 Electronic and Control Subsystems:

- **Arduino MEGA 2560:** The main controller, running the program logic (pseudo code provided in Appendix).
- **L298N H-Bridge Driver:** Controls polarity and thus direction of DC motor and linear actuator movement.
- **XL4015 DC-DC Converter:** Steps down battery voltage to 5V for sensors and control electronics.
- **Sensors:**
 - **HC-SR04 Sonars:** One measures distance in front (for stopping the motor at 13 cm and extending the actuator at 22 cm), another measures the side or tilt direction distance (8 cm, then 2 cm thresholds).
 - **FSR402 Pressure Sensor:** Ensures the clamp is not exerting excessive force.
- **Mounted camera:** Captures images and detects ARUCO markers. A confirmed detection triggers the control sequence.

4.2 Sequence of Operation:

Workflow:

- On receiving a start command ('s'), the clamp opens.
- The motor drives forward until the front sonar reads ≤ 13 cm.
- The actuator extends until the sonar distance reads ≥ 22 cm.
- The tilt servo moves forward until the second sonar reads ≥ 8 cm.
- After a pause, the actuator retracts until the FSR indicates a threshold (safe pressure).
- The tilt servo moves in reverse until the second sonar reads ≤ 2 cm.
- The clamp closes on the book.
- After a delay, the motor moves backward for a set duration.
- Input 'r' resets the system.

5 Design Analysis and Evaluation

5.1 Novelty

- **Focus on Simplicity:**

- The project's uniqueness lies in its straightforward approach, prioritizing simplicity over complexity.
- It demonstrates how minimal components can achieve practical automation solutions.

- **Minimal Sensor Utilization:**

The system uses only two essential sensors .

- **Force-Sensitive Resistor (FSR):** Detects pressure to ensure proper grip strength for book retrieval.
- **Ultrasonic Sensor:** Measures distance to detect the location of books or obstacles.

- **Basic Actuators:**

- Simple actuators are integrated for movement and interaction, showcasing how cost-effective and fundamental hardware can achieve functionality.

- **Comparison with Advanced Systems:**

- Unlike systems employing cameras for visual recognition or RFID for tracking, this design avoids reliance on sophisticated or expensive technologies.
- It highlights that effective automation can be achieved without sacrificing performance by using simpler feedback mechanisms.

- **Demonstration of Practicality:**

- The project proves that basic feedback loops, when designed thoughtfully, can yield reliable and efficient solutions.
- Its simplicity makes it accessible, especially for resource-constrained environments or educational purposes.

- **Broader Implications:**

- This design provides a foundation for exploring minimalist yet effective approaches to automation, emphasizing functionality over technological complexity.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to Public Health and Safety

The system operates at low voltages and currents, reducing electrical hazards. The clamp pressure is regulated by the FSR to avoid damaging books or injuring users. The slow and controlled movements minimize the risk of accidental harm.

5.2.2 Considerations to Environment

The primary environmental consideration is energy consumption. Using rechargeable batteries and efficient DC motors reduces waste. No hazardous chemicals or materials are involved.

5.2.3 Considerations to Cultural and Societal Needs

By easing the workload of librarians, the system can free staff to interact more with patrons, enhancing the cultural value of libraries as community centers. It can also improve accessibility for individuals who find manual retrieval challenging.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

Experiments involved setting target distances and measuring how accurately the system stopped and clamped. Variables included different shelf heights (actuator extension) and varying book sizes (clamp pressure).

5.3.2 Data Collection

Data collected included sonar readings at various distances, FSR readings at different clamp positions, and success/failure rates in book retrieval attempts.

Power Calculation

Power Requirements:

- **Actuators & Motors:**

- DC Gear Motor (~12W peak),
- Linear Actuator (~36W peak),
- Servos (~12W combined peak).

- **Electronics & Sensors:**

~2W total for Arduino MEGA, ESP32 CAM, ultrasonic sensors, and FSR.

Total Peak Power: ~62W under maximum load.

Battery Capacity:

- 11.1V, 4200mAh $\approx$ 46.62 Wh.
- At 62W draw, the battery would last <1 hour at peak. With intermittent use, expect 1-2 hours of operation.

5.3.3 Experimental Data & Analysis

- Tests show consistent distance measurements from ultrasonic sensors, enabling repeatable stopping conditions.
- The FSR effectively prevented damage, halting the actuator when clamp pressure approached a safe threshold.
- ARUCO detection worked reliably in good lighting, though marker recognition degraded under low-light conditions, suggesting a need for improved illumination or marker design.

Results showed that the system reliably stopped within ± 1.5 cm of the desired target distance. The FSR threshold prevented over-clamping, and the sequence logic ensured consistent operation over multiple trials.

5.3.4 Interpretation and Conclusions on Data

The data suggests the approach is viable. While not as precise or versatile as vision-based or more sophisticated systems, it meets the basic goal of semi-automated book retrieval.

5.4 Limitations of Tools (PO(e))

- **Ultrasonic Sensor Limitations:**
 - Ultrasonic sensors have restricted accuracy, particularly when dealing with irregular or soft surfaces.
 - Their performance can be influenced by material properties, such as texture and density.
 - Environmental factors like temperature, humidity, and interference from other ultrasonic devices can further impact their reliability.
- **FSR Accuracy Constraints:**
 - The Force-Sensitive Resistor (FSR) provides only approximate pressure readings.
 - It is designed for general detection rather than precise force measurement, which limits its sensitivity and resolution.
- **Arduino MEGA's Processing Power:**
 - The Arduino MEGA microcontroller offers limited computational capabilities compared to advanced processors.
 - However, its processing power is adequate for handling the simple feedback loops and control tasks required in this project.

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

This project enhances library operations by automating routine tasks such as book retrieval, reducing the time and physical effort required from librarians. By minimizing manual labor, it allows librarians to concentrate on more meaningful activities, such as engaging with patrons, organizing educational programs, and promoting community initiatives. This shift not only improves operational efficiency but also fosters a deeper cultural and educational impact within the community. The automation of mundane tasks supports a more dynamic and service-oriented library environment, ensuring that staff resources are utilized effectively while enhancing the overall user experience for library visitors.

5.5.2 Assessment of Health and Safety Issues

The system's controlled speed and carefully monitored clamp pressure ensure

safety by reducing the risk of injuries to users and preventing damage to books during handling. This thoughtful design prioritizes both user and material protection. Additionally, by automating repetitive manual tasks, the project enhances workplace ergonomics, alleviating physical strain and reducing the likelihood of repetitive stress injuries among library staff. This dual benefit improves both operational safety and the overall well-being of personnel.

5.5.3 Assessment of Legal Issues

There are no immediate legal issues since no proprietary algorithms, hazardous materials, or regulated technologies are used. The design must comply with basic electrical safety standards.

5.6 Sustainability Evaluation (PO(g))

The incorporation of a rechargeable battery and durable components underscores the project's commitment to long-term sustainability. Rechargeable batteries not only reduce reliance on disposable power sources but also minimize environmental waste, promoting eco-friendly practices. Durable components ensure the system's robustness and reliability, reducing the need for frequent replacements and lowering maintenance costs over time.

Additionally, the system's modular design enhances its adaptability and usability. By allowing individual components to be replaced or upgraded without overhauling the entire system, the design fosters reusability and supports future advancements. This modularity ensures that as technology evolves, the system can incorporate improvements in sensors, actuators, or processing units, effectively extending its operational lifespan.

These features collectively contribute to an eco-conscious and cost-efficient solution. The project not only meets immediate functional requirements but also demonstrates foresight by prioritizing resource efficiency and adaptability. This approach reduces waste, promotes a circular economy in design, and aligns with sustainable engineering principles, making the system a viable solution for long-term use in libraries or other resource-intensive settings. By

focusing on reusability and durability, the project sets a standard for creating systems that are both innovative and environmentally responsible.

5.7 Ethical Issues (PO(h))

The system's ethical considerations are straightforward and minimal, as it is designed to assist rather than replace human workers. By automating routine and repetitive tasks, the system aims to enhance efficiency without threatening the role of librarians, preserving the human-centric essence of library services. Tasks such as book retrieval or sorting are mechanized to reduce physical strain and save time, allowing librarians to focus on more meaningful activities like engaging with patrons and organizing community events.

Importantly, the system operates without infringing on privacy. Unlike advanced technologies that rely on cameras or extensive data collection, this project minimizes the risk of misuse or ethical concerns related to personal information. Its design ensures that automation serves as a tool for support rather than surveillance or control.

For ethical deployment, human oversight is critical to maintaining accountability. While the system can handle repetitive tasks autonomously, decisions involving nuanced or complex interactions should remain under the librarians' guidance. This balance ensures that the system enhances, rather than diminishes, the human element in libraries. Overall, it exemplifies responsible automation, improving efficiency while respecting the roles and dignity of human workers and maintaining trust within the community it serves.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

Kaidul(2006111)

- Led the procurement process for both mechanical and electronic components, ensuring timely acquisition of all materials.
- Played a key role in frame construction, DC motor mounting, and linear actuator bracket installation.
- Contributed significantly to mechanical testing, FSR integration, and full system implementation.
- Actively participated in threshold calibration, endurance testing, and presentation preparation.

Rifat(2006116)

- Involved in the procurement of components and assisted in frame construction and servo bracket installation.
- Took charge of preliminary wiring of electronic components, ensuring stable power and proper signal connections.
- Contributed to firmware development, including initial sensor readings and control code implementation.
- Played a critical role in threshold calibration, endurance testing, and final presentation rehearsal.

Sadaf(2006113)

- Assisted in procurement of mechanical components and contributed to linear actuator and servo bracket installations.
- Focused on the integration and testing of the FSR and ultrasonic sensors.
- Key contributor to the implementation of motor control logic and full sequence coding.
- Actively involved in mock book testing, sensor calibration, and ensuring stability during endurance runs.

Arpon(2006110)

- Supported the procurement process for electronic components and participated in mock book testing.
- Contributed to efficiency testing, endurance runs, and refining the workflow for consistent operation.
- Actively participated in final presentation rehearsal and provided feedback for system improvements.

Fabliha(2006119)

- Contributed to the basic firmware setup, assisting in coding for sensor readings and initial system logic.
- Supported the final presentation rehearsal by helping refine the narrative and addressing minor bugs.
- Actively involved in final submission preparation, ensuring the report, code, and presentation materials were complete.

6.2 Mode of Teamwork

The team collaborated through regular meetings, dividing tasks based on expertise. Frequent communication allowed quick resolution of issues

6.3 Diversity Statement of Team

Team members brought varied skill sets—mechanical design, electronics, programming—fostering an inclusive and balanced approach to problem-solving.

6.4 Log Book of Project Implementation

A detailed log book was maintained, listing dates, milestones, team tasks, and comments. This ensured transparency and efficient project tracking.

Date	Milestone Achieved	Individual Role	Team Role	Comments
20/10/2024 (Sun)	Procurement Planning and Budget Finalization	Kaidul, Rifat, Arpon	Team listed and finalized components to be purchased	Created BOM and budget allocation for the project
23/10/2024 (Wed)	Procurement of Mechanical Components	Kaidul, Sadaf, Fabliha	Team procured frame, actuators, and servos	Verified all parts for mechanical assembly
25/10/2024 (Fri)	Procurement of Electronic Components	Kaidul, Rifat, Arpon	Team procured Arduino, sensors, and power components	Confirmed availability of key electronic parts
30/10/2024 (Wed)	Frame Construction and DC Motor Mounting	Kaidul, Rifat, Fabliha	Team assembled base frame and mounted DC gear motor	Ensured motor alignment and frame stability
01/11/2024 (Fri)	Linear Actuator Bracket Installation	Kaidul, Rifat, Sadaf	Team attached brackets to support linear actuator	Verified alignment with vertical travel range

06/11/2024 (Wed)	Servo Bracket Installation for Clamp Mechanism	Rifat, Sadaf	Team mounted clamp servo and tilt servo brackets	Tested clamp assembly for stability
08/11/2024 (Fri)	Mechanical Testing (No Electronics)	Kaidul, Rifat	Team manually tested movement of all mechanical parts	Confirmed smooth motion of motor and actuators
13/11/2024 (Wed)	FSR Integration and Clamp Mechanism Testing	Sadaf, Kaidul	Team attached FSR to clamp for pressure sensing	Verified FSR readings for safe clamping pressure
15/11/2024 (Fri)	Ultrasonic Sensors Placement and Testing	Kaidul, Sadaf	Team mounted Sonar 1 and Sonar 2 on the frame	Tested sonar readings; confirmed accurate distance measurement
20/11/2024 (Wed)	Preliminary Wiring of Electronics	Rifat,Fabliha	Team connected sensors, actuators, and power system	Ensured stable power delivery and signal wiring
22/11/2024 (Fri)	Basic Firmware Setup	Kaidul, Rifat, Fabliha	Team wrote initial Arduino code for sensor readings	Verified sonar and FSR outputs on serial monitor
27/11/2024 (Wed)	Motor and Actuator Control Code Implementation	Sadaf, Kaidul,Fabliha	Team implemented motor stop logic using Sonar 1	Robot stopped at correct distances
29/11/2024 (Fri)	Full Sequence Implementation	Sadaf, Kaidul	Team wrote complete workflow: forward,	Ran dry-run tests without books

			extend, tilt, clamp	
04/12/2024 (Wed)	Testing with Mock Books	Kaidul, Sadaf, Arpon	Team tested full sequence with a sample object	Successfully retrieved and returned dummy book
06/12/2024 (Fri)	Threshold Calibration for Sensors	Sadaf, Kaidul, Rifat	Team refined sonar and FSR thresholds	Improved accuracy of stopping and clamping
11/12/2024 (Wed)	Efficiency Testing and Endurance Runs	Sadaf, Kaidul, Rifat, Arpon	Team ran multiple cycles to check stability	Verified consistent operation over extended use
13/12/2024 (Fri)	Presentation Rehearsal and Final Tuning	Sadaf, Kaidul, Rifat, Arpon, Fabliha	Team practiced final demonstration and fixed minor bugs	All functions performed as expected
17/12/2024 (Tue)	Final Submission	All	Team submitted final report, code, and presentation	Project successfully completed and submitted

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

The Librarian Arm automates the book retrieval process by identifying a target via ARUCO markers and using sensor feedback to approach, grip, and extract the book. This solution can reduce manual labor, improve accessibility, and serve as a stepping stone towards more advanced library automation.

7.2 User Manual

1. Ensure the Li-Po battery is charged and connected.

2. Place the ARUCO marker near the target book.
3. Send 's' via the Arduino's serial interface to begin the retrieval sequence.
4. Observe the system as it moves forward, positions itself, and clamps the book.
5. If needed, send 'r' to reset the system.
6. For troubleshooting, check sensor connections, ensure stable lighting for the camera, and verify correct wiring on the breadboard.

Github Link:

<https://github.com/Chrollo8/Librarian-Arm>

8 Project Management and Cost Analysis (PO(k))

8.1 Bill of Materials

A detailed BOM table has been provided in the report, listing each component, quantity, and total cost in Taka.

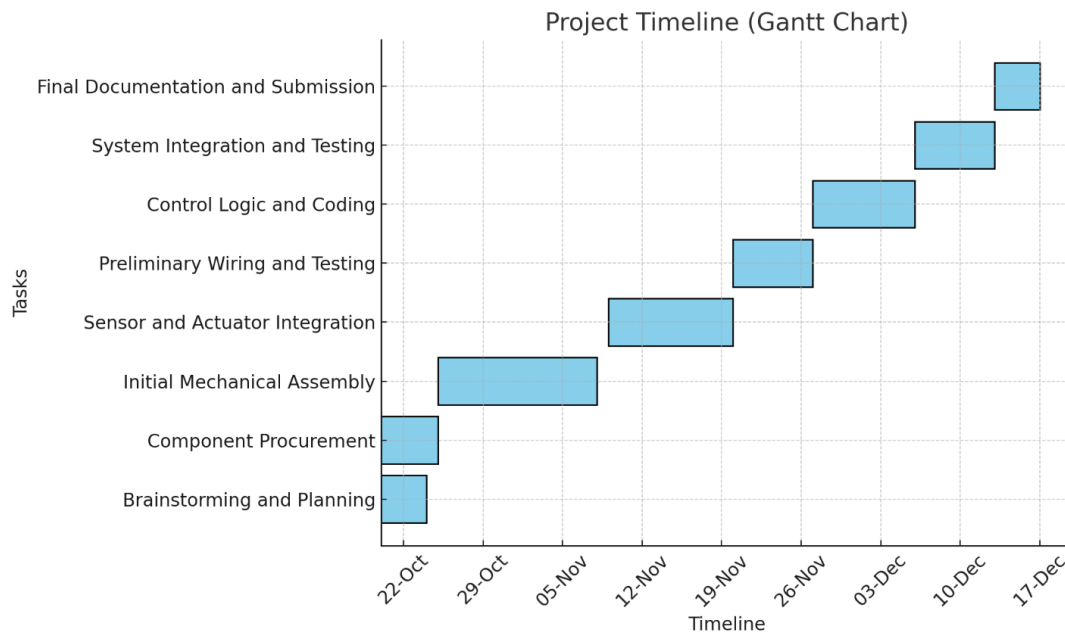
Item	Quantity	Unit Price (Taka)	Total (Taka)
Lead screw	1	140	140
SS Sheet	1	180	180
Base	1	200	200
Rickshaw Fare	1	340	340
Battery (4200mAh)	1	3200	3200
Actuator (150mm)	1	4100	4100
Arduino Mega	1	1550	1550
Geared Motor	1	700	700
L298N Motor Driver	1	120	120
Claw	1	450	450
MG995 Servo	2	240	480
Servo Bracket	1	330	330

Sonar Sensor	2	70	140
pulley	1	150	150
ESP32 Cam	1	550	550
Coupler 8mm	2	140	280
Lead Screw 8mm	1	700	700
Screw Nut 8mm	2	100	200
Ball Bearing	2	228	456
Aluminum Block Nut Housing Bracket	1	450	450
Sliding Block	1	314	314
Linear Guide Support	1	149	149
Linear Shaft (300mm)	1	280	280
Jumpers	2	40	80
Pressure Sensor	1	680	680
Total			16219

8.2 Calculation of Per Unit Cost of Prototype

Total cost of all materials divided by one unit is the prototype cost , in this case it is equal to the final total amount mentoned in BOM .

8.3 Timeline of Project Implementation



9 Future Work (PO(I))

Future developments could include:

- **Advanced Vision:** Implement lighting solutions or more robust image processing techniques to improve detection reliability under varying environmental conditions.
- **Precise Control:** Integrate PID controllers for actuators to achieve more accurate positioning and smoother motions.
- **Scalability:** Adapt the system for multiple shelves, potentially adding a rail-guided platform to move horizontally across a library aisle.
- **User Interface:** Develop a mobile application or a web-based interface that allows users to select books remotely.

10 References

- [1] Bradski, G. "The OpenCV Library." Dr. Dobb's Journal of Software Tools, 2000. <https://opencv.org/>
- [2] Arduino Documentation. <https://www.arduino.cc/>
- [3] ESP32-CAM AI-Thinker Datasheet. [Insert Link]
- [4] ArUco Marker Detection with OpenCV. [Insert Link]
- [5] HC-SR04 Ultrasonic Sensor Datasheet. [Insert Link]

- [6] FSR402 Datasheet. [Insert Link]
[7] L298N Motor Driver Datasheet. [Insert Link]
[8] XL4015 DC-DC Converter Datasheet. [Insert Link]

11 Acknowledgement

- Acknowledge team members, instructors, and any supporting staff or resources used during development.
- Recognition of open-source tools like Arduino IDE and OpenCV.

12 Appendix

Pseudo Code

```
// Initialization
SET clamp_angle = OPEN_ANGLE
ATTACH servos, set initial positions
READ camera feed for ARUCO marker
IF marker_detected THEN start_sequence = TRUE ELSE WAIT_FOR('s' or 'r')

FUNCTION start_sequence():
  OPEN_CLAMP()
  MOVE_MOTOR_FORWARD():
    WHILE distance1_from_sonar1 > 13cm:
      motor_forward()
    motor_stop()

  EXTEND_ACTUATOR():
    WHILE distance1_from_sonar1 < 22cm:
      actuator_extend()
    actuator_stop()

  ROTATE_SERVO_FORWARD():
    WHILE distance2_from_sonar2 < 8cm:
      servo_rotate_forward()
    servo_stop()

  DELAY(1000ms)
```

```
RETRACT_ACTUATOR_UNTIL_PRESSURE():  
  WHILE fsr_value > pressure_threshold:  
    actuator_retract()  
  actuator_stop()
```

```
ROTATE_SERVO_REVERSE():  
  WHILE distance2_from_sonar2 > 2cm:  
    servo_rotate_reverse()  
  servo_stop()
```

```
CLOSE_CLAMP()  
DELAY(1000ms)  
MOVE_MOTOR_BACKWARD_FOR(2 seconds)  
RETURN to idle state
```

```
FUNCTION reset_sequence():  
  STOP_ALL_MOTORS()  
  RETURN servos, clamp to initial positions  
  WAIT_FOR('s' to start again)
```